

MITHILESH A

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PROFESSIONAL SUMMARY

High-achieving AI & Data Science student at SASTRA Deemed University specializing in **Reinforcement Learning (RL)**, **Deep Learning**, and **Quantitative Finance**. Proven researcher achieving **95%+ accuracy** in financial forecasting models and developing intelligent urban infrastructure for the Smart India Hackathon (SIH). Adept at building full-stack web solutions integrated with advanced neural architectures.

EDUCATION

SASTRA Deemed University

B.Tech in Computer Science (AI & Data Science)

Thanjavur, TN

Aug. 2024 – June 2028

- **Focus:** Reinforcement Learning, Multi-Agent Systems, and Algorithmic Trading.

Harvard University (CS50)

Web Programming with Python and JavaScript (CS50W)

Remote

In Progress

- Mastering the Python stack, including Django, React, and SQL database integration.

RESEARCH EXPERIENCE

Student Researcher (Financial Markets)

DNN-RL Based Solution for Predicting Exchange Rates

SASTRA Deemed University

Mar. 2026 – Present

- Authoring a specialized quantitative research paper targeting **IEEE** publication.
- Integrating Deep Learning and policy-gradient Reinforcement Learning (PPO) to augment time series models.
- Incorporated critical macroeconomic variables including Crude Oil prices, Gold futures, and the NIFTY 50 index to achieve a predictive accuracy exceeding **95%**.

Student Researcher (Cybersecurity)

Advanced Persistent Threat (APT) Detection

SASTRA Deemed University

Feb. 2026 – Present

- Investigating APT detection methodologies and anomaly identification using Machine Learning to enhance network security frameworks.
- Leveraging Deep Learning (DL) models to comprehensively analyze network traffic and accurately detect distinct stages of vulnerabilities and cyber intrusions.

TECHNICAL PROJECTS

RL-Based Traffic Management (SIH 2025) | DQN, MARL, OpenCV, Raspberry Pi

Sep. 2025 – Dec. 2025

- Developing an intelligent urban signal control system for the Smart India Hackathon (Odisha) to minimize traffic clearance time.
- Integrated OpenCV on Raspberry Pi for real-time vehicle detection, preprocessing, and signal adjustments.

Web-Based AI Applications | Python, Django, MongoDB, ML

Jan. 2026 – Present

- Engineered various full-stack web platforms integrating custom AI/ML-backed solutions for seamless user interaction.

TECHNICAL SKILLS

Programming: Python (Advanced), Java, C++, C, SQL, JavaScript

AI/ML/Data: Deep Learning, Reinforcement Learning (PPO, DQN, MARL), TensorFlow, OpenCV, NumPy, Pandas

Web/Tools: Python Stack (Django), HTML/CSS, Git, MongoDB, Feature Engineering, Data Cleaning

Domain Knowledge: High Performance Computing (HPC), Quantitative Finance, Macroeconomic Analysis, DSA

CERTIFICATIONS & ACHIEVEMENTS

- **NPTEL SWAYAM:** Completed **Deep Learning Specialization**, mastering neural network architectures, back-propagation, and optimization (2026).
- **CMI STEMS:** Selected in the **Top 30 in India** for research aptitude by Chennai Mathematical Institute (2025).
- **Yale University:** Completed *Financial Markets* certification via Coursera, focusing on market theory.
- **Cisco Networking Academy:** Certified in Data Science, Python, and statistical analysis.
- **Mathematics Olympiad:** Merit certificate holder in the Indian Olympiad Qualifier in Mathematics (IOQM).